

End-to-End Geoacoustic Inversion With Neural Networks in Shallow Water Using a Single Hydrophone

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Abstract—This article presents a deep learning (DL) method to perform joint source detection and environmental inversion of low-frequency dispersed impulse signals recorded on a single hydrophone, in a fully automated way, with the inversion part covering both source localization (range and depth) and geoacoustic inversion (with the seabed modeled as a single sediment layer over a basement). The benchmark used for testing the resulting DL models are signals that were generated by navy explosives [signal underwater sound (SUS) charges] deployed during the Seabed Characterization Experiment 2022 performed in the New England Mud-patch (NEMP) off the coast of Massachusetts. A DL model based on a 1-D convolutional neural network is trained using simulated data. The resulting model is used to automatically process 816 h of acoustic data containing 289 SUS events. All the SUS events are detected (with no false positives), localized with a mean error of 400 m, and used to invert for seafloor geoacoustic parameters. The predicted parameters are in agreement with results obtained using classical inversion schemes. Using a trained DL model requires little to no computation time and power, compared to classical methods, which employ high-cost computational schemes. This advantage enables efficient inversion of enough SUS events (289) to spatially cover the NEMP, and inversion results suggest spatial variability in the mud sound speed.

Index Terms—Deep learning, geoacoustic inversion, New England mud patch, SBCEX, seabed characterization experiment, underwater acoustics.

I. INTRODUCTION

UNDERSTANDING how acoustic waves propagate in the ocean is of utmost importance in acoustical oceanography [1]. This knowledge can be of particular interest in a Navy context, but more recently has become an essential component in marine science research. For example, understanding the effects of anthropogenic sound pollution on marine ecosystems requires accurate simulations of acoustic propagation in the ocean. This article, focuses mainly on coastal environments (depths less than

200 m), where the sound field is highly affected by sea surface and seafloor interactions.

This bounded medium acts as a dispersive waveguide, and normal mode theory [2, Ch. 5] can be used to describe the acoustic field accurately. Good knowledge of the seafloor's geoacoustic properties is crucial in this environment. But, direct measurements of geoacoustic parameters are difficult and expensive. Thus, indirect methods are used, namely, extracting embedded information in acoustic signals that propagated in an environment of interest. The indirect extraction of geoacoustic properties from acoustic signals is called *geoacoustic inversion*. Sometimes, environmental inversion problems are merged with source localization. This approach is applied in this work.

Seeking solutions to inverse problems requires high computational power, as demanding algorithms, such as the Markov Chain and Monte Carlo (MCMC) methods are used, and thus, their application in real time is not a feasible prospect. To solve this problem, recent efforts focus on improving further the inverse problem solution schemes and current methods for solving the forward problems (propagation models) that are called upon when seeking the inverse solution [3].

An alternative to these traditional inverse methods is available in machine learning (ML), and more recently deep learning (DL), which make use of neural networks (NN). ML and DL, the most recent revolutions in computer science, result from the introduction of parallel computing using graphical processing units (GPUs). Recent ML and DL methods have been applied to the development of popular artificial intelligence algorithms and to answer scientific questions faster, sometimes more accurately [4]. The impact of DL methods on the characterization of underwater environments using acoustic sources stays limited, due to two main reasons. First, large labeled datasets are usually not available: a crucial component for a successful DL implementation. Second, the signal degradation due to underwater propagation and noise is more severe than for other classical ML applications. To tackle the issue of insufficient available data, physical models can be used to produce synthetic data samples to complement this deficiency. Using these models, we can simulate the acoustic fields in diverse environments and train a NN that can predict geoacoustic properties using real acoustic signals.

By taking this approach, an NN is optimized to act as a solver for the inverse problem in a particular environment, instead of using optimization methods to arrive at a solution directly, or

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via sampling in the case of Bayesian inversion. Using a trained NN requires only the algebraic multiplication of tensors and can be performed using low-powered computer chips, usually found in personal computers. Thus, real-time on-site solutions can be sought where access to a powerful computer is limited. DL methods have already been proven to perform well in geoscience, showing substantial performance increases in solving both the forward and inverse problems [5].

The first attempt to use ML for geoacoustic inversion was as early as 1996 by Caiti et al. [6], where radial basis functions were used to approximate the inverse mapping of acoustic signals to geoacoustic parameters. In the late 90s, linear feed-forward NNs were used by Stephan et al. [7] and Benson et al. [8] for geoacoustic inversion, and promising results were shown over synthetic and experimental data. Even though successful, these methods were limited by the lack of computational power at the time and access to more modern network architectures used today. More recently, several achievements were reached using trained NNs, such as source localization in the China Yellow Sea using a single hydrophone by Niu et al. [9], performing seabed classification in concert with source localization of navy explosives using a single hydrophone by Van Komen et al. [10] and Howarth et al. [11], direct source localization by Weiss et al. [12] and source localization using a data-driven approach that uses ship noise by Durofchalk et al. [13]. Sources of opportunity, such as moving ships, were also used for seabed classification using DL by Neilsen et al. [14] and Escobar-Amado et al. [15]. Further, ML was also used to solve the forward problem by Kexin and Chitre [16], Varon and Bonnel [17], which can aid decreasing the computational cost needed to solve the inverse problem using traditional methods.

This article demonstrates a similar approach to [10] and [11], but differs from them by performing a regression task for geoacoustic parameters inversion instead of a seabed-type classification task, using a fully automated pipeline. Similarly, as done in the pioneer works [6], [7], and [8], the objective is to obtain results that are fully equivalent to those obtained with traditional inversion methods (e.g., parameter estimation problems [3], [18]), while making the inversion process automatic and computationally lighter.

The objective of this article is to present a fully automated DL approach to the characterization of the underwater environment using acoustic sources, specifically navy explosives called signal underwater sound charges (SUS) [19]. Recordings of such explosives were recently taken in multiple experiments during the seabed characterization experiments (SBCEX). We use SUS recordings collected in May 2022 (SBCEX22) aboard multiple single hydrophones moored on the seafloor at different locations and extract from them the geoacoustic parameters of the propagation environment using a trained NN. This process includes automatic SUS event detection, source localization, and environmental characterization without any manual intervention. The results of this work are used to validate the performance of the developed DL approach and are compared to other results obtained using traditional methods on the same experimental site. Note that the proposed approach is similar to the work of Goldwater et al. [20] which uses a single hydrophone and DL to automatically detect and range dispersed impulsive signals from

baleen whales. A different data pipeline and NN architecture is used here. More importantly, the inversion scheme is extended to cover both source depth and environmental geoacoustic parameters.

The rest of this article is organized as follows. Section II describes the SBCEX22 experiment and the experimental data that is used for testing the trained NN for detection and inversion tasks. Section III goes over the methodology for training both a detection and inversion NN, which includes the data generation process. Section IV showcases the DL-based inversion results on SBCEX22 data and evaluates the localization and detection performance as well. Finally, Section V concludes this article.

II. BENCHMARK: SBCEX22

To test the performance of a trained NN, we use as a benchmark the SBCEX22 experiment, performed in the New England Mud Patch (NEMP) in May 2022. An overview of the experimental site can be found in [21] and [22].¹ The NEMP has been heavily surveyed in the last decade, and, thus, it enables us to have independent knowledge about the underwater environment, which can be used to validate our inversion results and performance of the trained NN. In this work, the data were collected by 17 TOSSITs; single hydrophones moored 1 m from the ocean bottom [23]. The data of interest for this article were collected between May 17 and May 19 (48 h), leading to a total of 816 h of acoustic data. This dataset includes recorded signals from 17 SUS deployments. Since all the SUS charges have been recorded on all the TOSSITs, a total of 289 impulsive signals are available in the dataset. The TOSSITs were positioned around the detonation coordinate at 40°27'72" N 70°33'44" W, with ranges of 2 to 9 km between hydrophones and the SUS event, as seen in Fig. 1(a). The nominal SUS detonation depth is around 18 m [24]. The experimental area's water depth is approximately constant, which allows the assumption that all acoustic signals propagate in a range-independent environment [see Fig. 1(a)]. Measurements of the water sound speed profile (SSP) were performed between 17 May–19 May at the locations in Fig. 1(a). The SSPs in Fig. 1(b) show a dynamic profile that varies in time and space.

III. METHODS

A. Synthetic Data Generation

Due to the scarcity of experimental training data, physical simulations are used to create a sufficiently large synthetic dataset for training. The signals are generated using KRAKEN [25], a *normal mode* propagation code. Each environment used for simulation is modeled using a one-sediment layer geoacoustic model, as illustrated in Fig. 2. All environments are generated using a parameter grid defined by geoacoustic parameters of interest for inversion. They are the sediment-layer thickness h_1 , the sound speeds of the sediment layer c_1 , and the bottom half-space c_B and the density of the sediment layer ρ_1 and the bottom half-space ρ_B . The bounds for each geoacoustic parameter, listed in Table I, are defined using knowledge of

¹No overview of the 2022 experiment has been published yet, but [21] and [22] provide an overview of the experiment in 2017 to the interested reader.

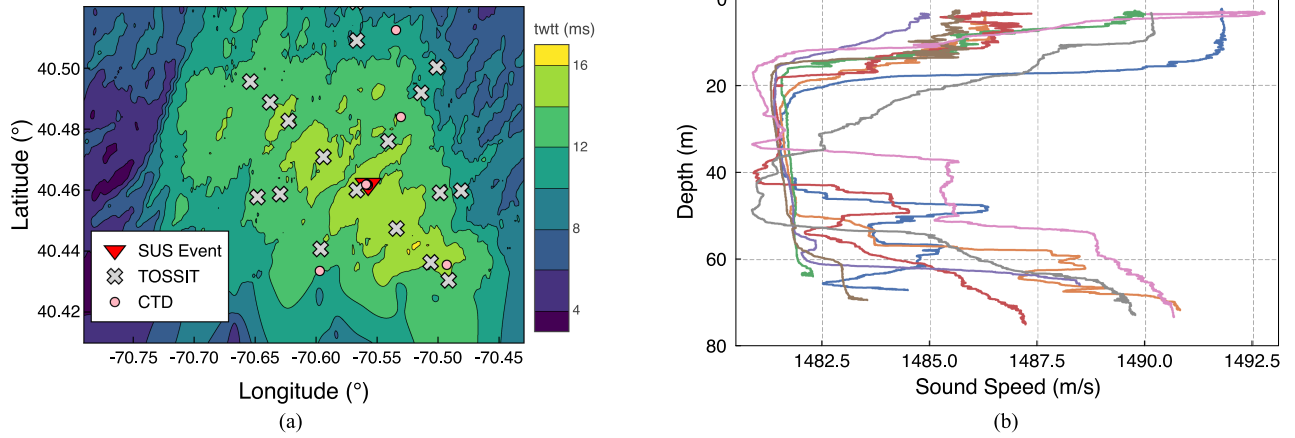


Fig. 1. (a) Two-way-time-travel and coordinates of the SBCEX22 experiment area. TOSSIT locations are labeled as red pointers, the SUS event location is labeled as a green pointer and CTD locations are in pink markers. Mud thickness is described using *two-way travel time* in ms. (b) Measured Sound speed profiles during the SBCEX22 experiment between 17-19 May 2022.

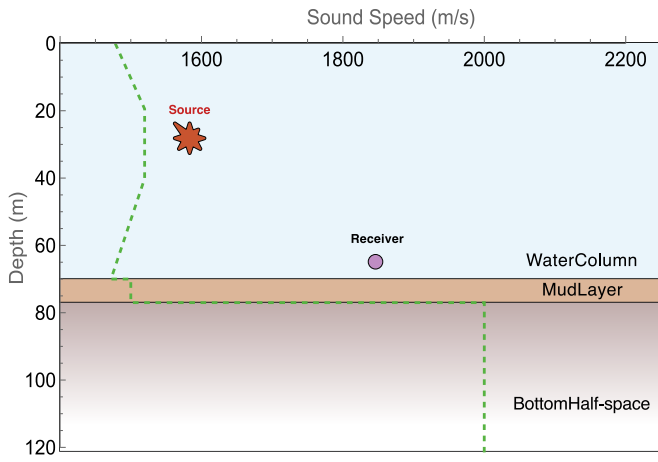


Fig. 2. Illustration of geoacoustic model used for synthetic dataset creation.

TABLE I
PARAMETERS USED FOR SYNTHETIC DATA GENERATION

Parameter	Unit	Bounds/Value	Step
<u>Water Column</u>			
No. of nodes	—	4	
Sound Speed At Nodes	m/s	[1478, 1497]	Random
Seafloor Depth h_0	m	[64, 75]	Random
<u>Mud Layer</u>			
Thickness h_1	m	[5, 15]	1
Sound Speed c_1	m/s	[1400, 1600]	15
Density ρ_1	g/cm ³	[1.2, 2]	0.1
Attenuation α_1	dB/λ	0.05	
<u>Bottom Half-Space</u>			
Sound Speed c_B	m/s	[1600, 2000]	40
Density ρ_B	g/cm ³	[1.8, 2.4]	0.1
Attenuation α_B	dB/λ	0.25	
<u>Sound Receiver</u>			
Source Depths z_s	m	[10, 40]	2
Source Range r	km	[0.5, 10]	0.1
Receiver Depth z_r	m	1 m from bottom	

the NEMP values derived from previous geoacoustic inversion results [22].

For each set of seafloor geoacoustic parameters, a random water sound speed profile (SSP) is generated. This SSP is described using four nodes at depth 0, 20, 40, and h_0 , with h_0 a random water depth between 64–75 m. The sound speed at each node is also randomly drawn between 1478–1497 m/s, and the full SSP is constructed with linear interpolation between the nodes. This process allows the data to account for the water SSP uncertainty due to the dynamic behavior of the water column during the SBCEX22 experiment. All environmental parameter values and bounds can be seen in Table I.

For each environment, KRAKEN calculates the appropriate modes for each frequency. The modes are then used to generate the synthetic spectra for the SUS charges using the following equation:

$$p(r, z, z_s; f) \cong S(f, z_s) \frac{ie^{-i\pi/4}}{\sqrt{8\pi\rho_w}} \times \sum_{m=1}^N \Psi_m(z_s, f) \Psi_m(z, f) \frac{e^{ik_{r,m}(f)r}}{\sqrt{rk_{r,m}(f)}} \quad (1)$$

where r is the range between the source and receiver, z is the receiver depth, z_s is the SUS detonation depth, f the frequency, ρ_w is the water density, Ψ_m the modal depth function with the associated modal wave number $k_{r,m}$, N is the number of calculated modes, and S is the source spectrum. The source spectrum for an SUS charge is modeled using [24], leading to the depth-dependent $S(f, z_s)$. An inverse fast Fourier transform (IFFT) of the complex spectrum yields a synthetic pressure time-series.

Each signal in the synthetic dataset is a 2-s pressure time-series, sampled at 1000 Hz. Therefore, KRAKEN is run for frequencies 1–500 Hz with steps of 0.5 Hz. The number of modes calculated is $N = 41$, which is the most required by an individual environment for the highest frequency used, which in this case is 500 Hz.

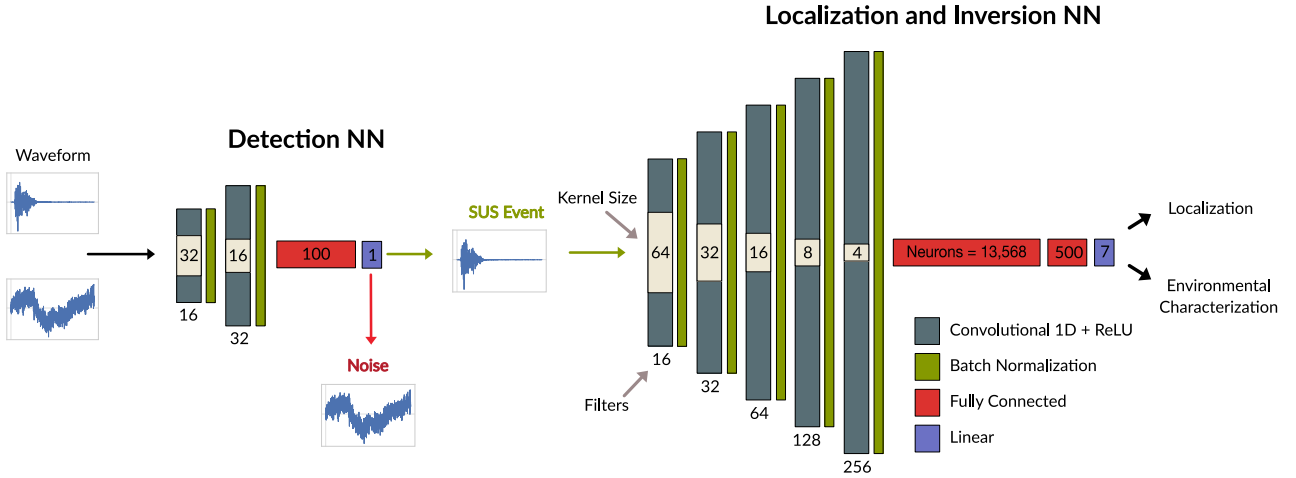


Fig. 3. Neural network architectures.

After an IFFT is taken, each time-domain signal is normalized by its maximum absolute amplitude. This normalization guarantees that any resulting output from the NN relies solely on the relative values of the signal. The beginning of the SUS event is defined as the first time the absolute relative amplitude of 0.1 appears at 0.1 s, and thus, each generated time-series is realigned accordingly.

In total, 25 000 environments are formed using the parameter grid (see Table I). For each environment, signals are synthesized using (1) with the range r varying from 0.5 to 10 km with 0.25-km steps, and the source depth z_s varying from 10 to 40 m with 2-m steps. By using all environments, ranges, and source depths, the resulting dataset consists of 4 118 400 synthetics pressure time-series, which we call the “*Big dataset*.” A smaller dataset is generated as well, using only 1000 randomly chosen environments using the same parameter grid, and consists of 164 736 synthetic pressure time-series, which we call the “*Small dataset*.” The *Small dataset* is included in the *Big dataset*, and is used to determine the ability of an NN to learn from a smaller dataset without needing to fully grid the parameter space. Each sample (i.e., time-series) is labeled by the parameters of interest, using a length-7 vector $[r, z_s, c_1, h_1, \rho_1, c_B, \rho_B]$. It is important to note that the water SSP parameters used for each environment are not included in this vector: they are seen as random parameters, and they will not be estimated by the NN.

B. Train Localization and Inversion NN

When training an NN, both synthetic datasets, *Big* and *Small*, are used. The automatic pipeline consists of two NNs chained together, one for detection and the other for concurrent localization and geoaoustic inversion. The input to both NNs is a normalized pressure time-series. The detection NN outputs a probability that the input signal is the signal of interest, and the second NN performs the inversion task on the signal of interest. We first describe the training of the second NN performing inversion, as this is the most intricate and complicated step, and then using a similar approach, we describe the training of a detection NN

The inversion NN is composed of a chain of five 1-D convolutional layers, and three linear layers. A batch-normalization layer is attached to every layer and helps stabilize the training process. Each convolutional layer is composed of several kernels of a specified length. The kernels in the layer are *convolved* over the input vector. In contrast to a more traditional convolution definition used in signal processing, convolution in an NN means calculating the weighted sum over an input vector using a kernel with steps of size *stride*. Here, the size of the *stride* is 2 for all convolutional layers in our NN. All layers are activated using a ramp function (generally called an ReLU function) except for the last output layer, which is not activated. The overall architecture is presented in Fig. 3 along with extra details on the number of kernels (filters) and their sizes in each convolutional layer and the length of each linear layer.

1D-CNN-based architectures are widely used for temporal and sequential data, and the network architecture has been inspired by work done in [26] for sound classification. This is because convolutional layers can learn local dependence and time structures of audio signals, similarly to 2-D convolutional layers when working with images. The layer’s kernels are learnable through the training process, and each kernel eventually learns to filter out specific features relevant to the NN task. *Depth* in this case is required, since NNs with simpler architectures were unable to adequately learn from the data and then perform geoaoustic inversion on synthetic test data. Through a trial-and-error process, a *complex enough* architecture was devised (inspired by Abdoli et al. [26]) to be able to learn and predict over synthetic data to a reasonable degree of success. Deeper networks (or more complex networks in general) could have performed better, but discovering the optimal NN architecture for geoaoustic inversion by NN hyperparameter optimization is out of the scope of this article. For more information on different types of layers and their mechanisms, see [27], and [28] for more specific detail on current uses of 1D-CNNs. The network takes as input a vector of length 2000 and outputs seven values corresponding to two localization values, r and z_s , and the five geoaoustic parameters of interest, described in Section III-A.

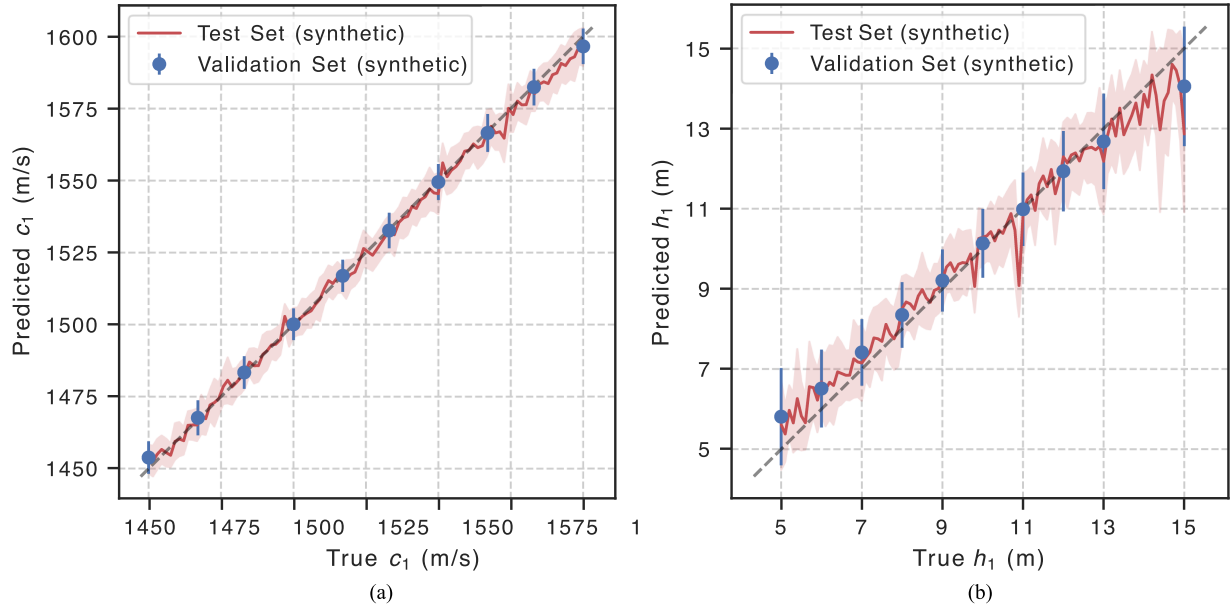


Fig. 4. True labels versus mean and STD of predicted labels for the synthetic validation data (blue) and synthetic test data (red) for (a) sound speed c_1 in the sediment layer (b) thickness h_1 of sediment layer. The gray dashed line represents a visual guide for a perfect match between true and predicted labels.

TABLE II
SUMMARY OF THE HYPERPARAMETERS USED FOR TRAINING THE
LOCALIZATION AND INVERSION NN

Parameter	Value
Loss	MSE
Optimizer	ADAM
Learning Rate	0.001
Batch Size	512
Max Epochs	100
Early Stopping	10 epochs (patience)

All the parameters and their associated values are discussed in Sec. III.B

TABLE III
MEAN ABSOLUTE ERROR (MAE) BETWEEN TRUE AND PREDICTED VALUE
USING SYNTHETIC VALIDATION AND TEST DATASETS

Parameter	Validation MAE	Test MAE
Range (km)	0.4	0.38
Source depth (m)	0.5	0.5
c_1 (m/s)	4.58	4.76
ρ_1 (m)	0.19	0.21
h_1 (kg/m ³)	0.79	0.76
c_B (m/s)	7.25	8.79
ρ_B (kg/m ³)	0.16	0.15

The *PyTorch* package, interfaced by the *Python* computer language, is the chosen DL framework for training the NN on the synthetic dataset using hyperparameters listed in Table II. A training round, called an *epoch*, consists of passing the entire training dataset through the NN. During each epoch, the training data are fed to the NN in batches of 512 data samples (signals). After each batch pass-through, the NN weights are modified by the optimizer such that the loss function is minimized over the difference between the predicted and expected outputs.

To each signal passing through the network, Gaussian noise is added using a randomly chosen SNR value in 10–30 dB. The SNR is calculated over the entire signal and is defined, in dB, as $10 \log_{10}(P_{\text{signal}}/P_{\text{noise}})$ where the power P is defined as the variance of the signal. A different random noise signal is added to each input signal every time it is passed through the network during training (i.e., each *epoch*).

The loss function is the mean square error (MSE) between the predictions made by the NN and the associated labels for each signal over the entire batch. The loss function is minimized over each batch by adjusting the NN filters and weights using an optimizer based on gradient descent. The optimizer of choice

is called adaptive moment estimation (ADAM), a widely used state-of-the-art optimizer [29], and it is used with a learning rate of 0.001.

During training, the NN performance is evaluated using a validation dataset at the end of each epoch. The overall validation step is performed over two different validation datasets. The first set is a synthetic validation set composed of a 0.01% subset of the entire synthetic dataset which is put aside and is not used for training. The second is an experimental validation set made of 15 randomly chosen signals out of the 289 recorded SUS events detected in the experimental recordings using the detection NN.

The *mean absolute error* (MAE) is calculated over the synthetic validation set to determine the ability of the NN to generalize to synthetic data after each full training round (Epoch). The MAE results calculated over the synthetic validation set at the end of the training process are presented in Table III. Furthermore, performance on the synthetic validation set is visualized by presenting predictions on the sound speed in the sediment layer c_1 and the sediment layer density h_1 in Fig. 4 (other parameters show similar behavior).

The *experimental* validation set is used to evaluate localization performance by measuring the MAE between the known source range [range between TOSSIT and SUS event, as shown in Fig. 1(a)] and the predicted range on the experimental test set. In addition, source depth and geoacoustic parameter estimation are predicted on that same set. For the NN predictions on the 15 measured data samples, mean and STD are calculated for each parameter estimation separately. Both the mean and STD on the experimental validation set are used to measure the ability of the NN to generalize to experimental data, as the NN does not see any experimental data during training. These measures are used since the *true* geoacoustic parameters associated with each measured signal is unknown. The training process is stopped once the STD of the predictions for c_1 reaches a local minimum value in the span of a specified number of epochs. The number of epochs lapsed until a local minimum is determined is called *patience*, and here was decided to be 10 epochs. This process of stopping training after a local minimum is found after a certain number of epochs is called *early stopping*, and is common practice in the DL community [30]. The STD criterion over c_1 gives an indication that the NN has converged to a stable solution for experimental data, but is not exclusive to that specific parameter. The maximum number of epochs allowed for training is 100 and in all our training, the NN converged to a stable solution, usually after 20–40 epochs. Differently from the STD, the mean values over the predictions are used only qualitatively as an assurance that the predicted geoacoustic parameters are not too far off the parameter space used to create the training data.

C. Train Detection NN

A detection NN is trained using a similar approach to Section III-B but with minor differences, mainly in the early stopping criterion and the dataset used. The detection model takes in a 2-s signal (of length 2000) and outputs a probability of the input signal being of interest (value between 0 and 1). In this article, the signal of interest is a recorded SUS charge event, and every other input is considered noise. If the predicted probability is above a certain threshold, which is predefined by the user, the signal is extracted and time-shifted to have the SUS event appear at a start time of 0.1 s in the time window (as was done in Section III-A when generating the synthetic dataset). We use 100 h of experimental noise from TOSSIT recordings, and 100 h of randomly picked SUS signals from the synthetic data generated previously to train the NN. The noise recordings were resampled to 1000 Hz, segmented into 2-s windows, and normalized to have a mean of zero and a maximum absolute value of one. All signals in the synthetic dataset represent an SUS event and are labeled with a probability of 1, and all signals in the experimental noise set are labeled with a probability of 0. The combined dataset is then split into a training set and a validation set, with a 99.9%–0.1% split.

The NN architecture used for detection is a shallower and smaller version of the inversion NN architecture, and can be seen in Fig. 3. From observations, any smaller network architecture was able to achieve very high performances, therefore, the

presented NN is one of many possible options. *Early Stopping* is used to stop training when the MSE on the validation set reaches a local minimum in the span of 10 epochs, similarly to the training process described in Section III-B.

A postprocessing step is performed to deal with multiple detections of the same SUS signal over multiple successive overlapping windows due to the overlapping segmentation of the long recording. Detections that are associated with the same SUS signal are grouped together, and a 2-s segment is cut out of the long detected window where the signal start time value is 0.1 at 0.1 s from the beginning of the window.

IV. RESULTS

A. Results on Simulation Data

Once the model is trained, performance is evaluated over a synthetic test dataset created using 1000 environments with randomly sampled geoacoustic parameters within bounds similar to the ones presented in Table I. The geoacoustic parameters used to generate the environments do not sit on the grid used for creating the training datasets (the *Big* and *Small*), but are picked randomly within the same bounds. This performance metric ensures that the model can generalize to unseen synthetic environments in the same bounds. The MAE is calculated between the true labels and the NN predictions made on the test dataset, and the results are presented in Table III. The results are also presented graphically in Fig. 4, giving a visual representation of the performance of the NN on the test dataset. For brevity, we show the visualization for only the parameters c_1 and h_1 , but similar results are obtained for the other parameters.

An interesting observation can be made by comparing the results for predicted labels on the synthetic test dataset and the results for the synthetic validation dataset. The performance on both sets is very similar, indicating that the NN can generalize just as well on the validation set (which is generated using environments on the parameter grid) as on the test set (which is generated using randomly sampled environments). A bias error can also be seen in Fig. 4(b). As the NN trains, the weights are modified such that the cost function is minimized over the training data, but the minimization is not perfect. This imperfection can manifest itself as a bias error, which likely happened for the predictions over both synthetic validation and test data for h_1 . Nevertheless, the resulting NN performance is a good indicator that the NN can generalize to unseen environments using parameters that lie within the parameter bounds used for training data generation.

B. Results on SBCEX22 Data

1) *Detection Performance*: All recordings from the TOSSITs taken between May 17–19, 2022 were segmented into 2-s windows with a 1.9-s overlap and then passed through the trained detection NN. Overall, 816 h of data were processed through the detection model. All signals that had a probability prediction larger than 0.9 were selected and considered to be SUS events. All 289 signals were recovered, with no false positives (noise classified as an SUS event). All detected SUS

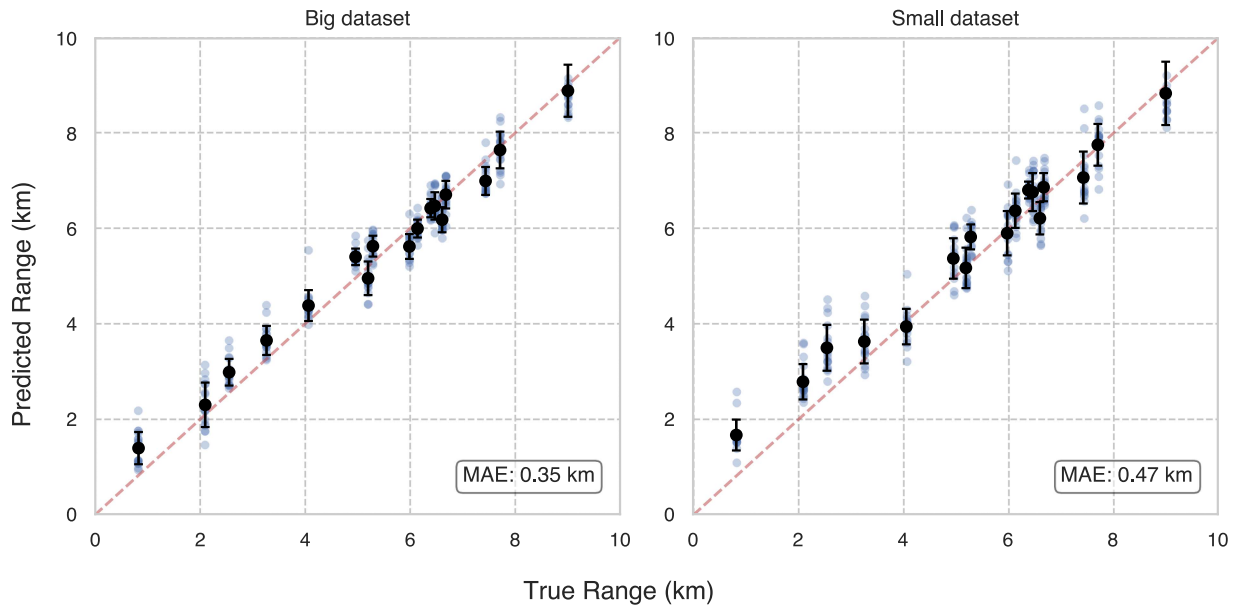


Fig. 5. Mean and STD of range predictions from hydrophone to SUS charges using NNs trained on synthetic data. Predictions are made using models trained on the *Big* dataset (left) and *Small* dataset. Individual predictions are shown as blue dots, with the mean & STD in black. The red dashed line acts as a visual guide for a perfect match between true and predicted labels. The mean absolute error (MAE) for all predictions is shown for each case.

events were realigned to have the event itself start at 0.1 s in the time window, as shown in Fig. 3.

The good detection results are not entirely surprising, as SUS events are strong, impulsive, and easily detectable with the human eye but also using detectors based on energy threshold. But a key difference is that the detection NN exclusively focuses on *dispersive* impulse signals, rather than any impulses. Thus, using a trained detection NN can help discern between subtle differences between signals of interest and noise, which can be useful in scenarios where the signals of interest are harder to find in very long noisy recordings. In practice here, the NN was able to reject various impulsive sounds that were either not loud enough or too close from the receivers to show dispersion.

2) *Inversion Performance*: The detected SUS signals were passed through the inversion NN, which was trained on both the big and small datasets. Two different performance metrics are evaluated on SBCEX22 data.

First, range predictions are compared to the known distances between the TOSSIT and the source. The mean and STD of the range predictions are calculated on signals recorded on each TOSSIT, and the results are presented in Fig. 5 for both trained NNs. The overall performance of each NN is measured as the MAE between the predicted and true range, which is 0.35 km when training on the *Big* dataset and 0.47 km when training on the *Small* dataset.

The second performance metric is the mean and STD of predictions from all signals over the source depth and all inversion parameters, and are presented in Table IV. To better understand the distribution of the predictions for every single parameter, a histogram of the predicted source depths and the inversion results are presented in Fig. 6 for both trained NN. When evaluating geoacoustic inversion performance, the STD is an indication of the *confidence* with which the NN is making its

TABLE IV
INVERSION RESULTS ON SBCEX22 DATA

Parameter	Small dataset		Big dataset		Bonnell et al. 2022	
	Mean	STD	Mean	STD	Mean	STD
c_1 (m/s)	1488.6	11.2	1486.6	8.4	1489.4	14.1
ρ_1 (kg/m ³)	1.64	0.05	1.64	0.04	1.57	0.07
h_1 (m)	8.8	1.5	8.9	1.4	8.5	1.6
c_B (m/s)	1785.6	37.2	1781.5	33.2	1849.0*	58.2
ρ_B (kg/m ³)	2.14	0.02	2.09	0.02	2.1	0.1

* In Bonnell et al., results were achieved on a variable number of sediment layers, thus, the bottom half-space sound speed value is not entirely comparable to our results which use a 1-layer sediment model.

predictions on each parameter. This *confidence* metric through the STD is only relevant due to the assumption of range-independent source-receiver tracks (constant geoacoustic parameters) and a constant SUS detonation depth.

To assess the reliability of our results, some comparison is made to published literature. According to [24], SUS charges are released when they reach a water depth of approximately 18 m, which correlates well with the results achieved with our trained NN. Moreover, our geoacoustic parameter inversion results are similar to what was published previously by Bonnell et al. in [31] when using Trans-D inversion on even modes, which is a good indication that our NNs can learn the underlying physics of the problem (see Table IV for a numerical comparison). It must be noted that the inversion technique used by [31] inverts for the number of layers as well, thus, the bottom half-space interface is deeper, and as a result, the bottom half-space sound speed is higher in [31] than our predicted sound speed.

An interesting observation can be made when comparing the results of the big and small datasets. The mean and STD of the inversion results are very similar, which is a good indication that

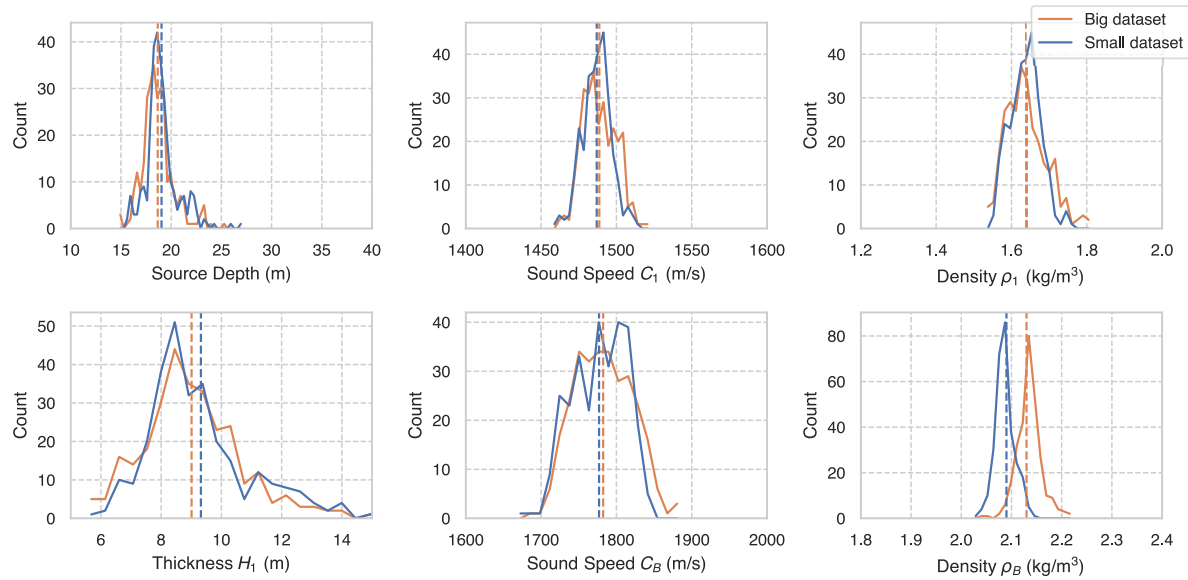


Fig. 6. Histogram of inversion results in counts using both models, with the mean prediction of each parameter shown as a vertical dashed line.

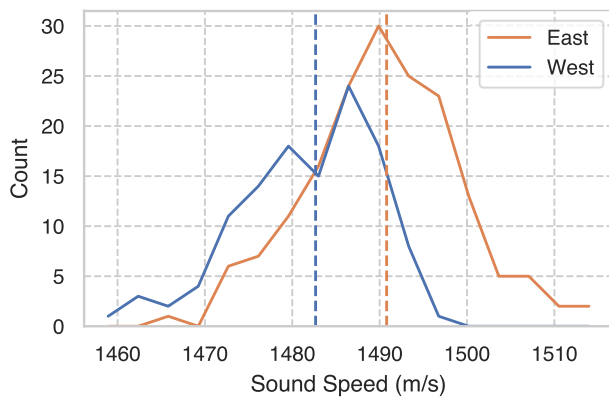


Fig. 7. Histogram of predictions made on the sediment sound speed as a function of west/east TOSSIT location with respect to the SUS event location, with the mean prediction shown as a vertical dashed line.

the NN trained on the small dataset can generalize to experimental data as well as the larger dataset. Therefore, smaller datasets can be used to train an NN that can be deployed in practice, as using this approach it is much faster to generate and train an NN.

V. CONCLUSION

Using a fully automated pipeline, we have been able to successfully detect and localize SUS charges in a shallow underwater context along with estimating the geoaoustic parameters of the environment, using DL methods and a single hydrophone. The results of the inversion are in good agreement with previous literature, and the NNs can generalize well to unseen data, both synthetic and experimental. The NNs can detect SUS charges with a high degree of confidence, and the localization results are consistent with experimental ground truth, and environmental inversion results are in agreement with what is known about the NEMP.

The method relies on two chained NNs, one for detection and one for inversion. Once both NNs have been trained, the computation time for the automated pipeline on each signal is negligible, so that large experimental datasets can be processed, and real-time applications can be tackled. As an example, applying a detection NN and an inversion NN on a 2-s signal takes about 0.01 and 3 ms, respectively, using a conventional consumer grade CPU found in personal computers. The runtime speed allows us to sift through a large amount of data, and perform localization and inversion tasks instantly. Furthermore, the models themselves take little storage space, each inversion NN model requiring 80 MB of hard-drive storage. Therefore, through-the-sensor environmental estimation methods, embedded into autonomous platforms, can be envisioned.

To invert for geoaoustic parameters describing a more complex geoaoustic model, such as a gradient in the sediment sound speed or several sediment layers, one would need to increase the number of outputs in the NN's last layer to accommodate the increase in inversion parameters. The training data must include signals generated using a bigger parameter space, and the depth or complexity of the NN might need to increase. But once the NN is trained, the inference time will still stay very low compared to traditional inversion methods.

In this article, the NN versatility was used to process 17 SUS charges deployed during SBCX22 and recorded on 17 TOSSITs, leading to 289 events of interest. This led to 289 new inversion results on the NEMP, something that would not be possible with traditional inversion methods. The diversity of all the SUS/TOSSIT combinations allows studying the spatial variability of the mud geoaoustic parameters.

In Fig. 7, the spatial variability of the sediment layer sound speed c_1 is presented as histograms of predictions made on signals captured by TOSSITs west and east of the SUS event coordinates (longitude -70.58°). The results indicate that there

is an overall increase in the sound speed, with the mean increasing from 1483 to 1491 m/s. These results are in agreement with recent coring results suggesting a constant increase in sound speed ratio from west to east in the NEMP [32]. Moreover, our results are in further agreement with inversion results obtained using a *Trans-D* inversion scheme applied to reflection (RC) data collected during the 2017 experiment. The RC inversion results obtained by Belcourt et al. [33] on the shallow-water acoustic measurement instrument (SWAMI) and the Vibracore Core 31-2 (VC31-2) site and results obtained by Jiang et al. [34] on the south-eastern site (SC2) also show an increase in the mud surficial sound speed from west to east.

Using the presented DL method showcased in this article, the gain in computation speed obtained using DL can be used to automatically process a relatively large amount of acoustic data, which in turn helps to better understand the spatial variability of the geoacoustic properties of the NEMP. The potential use of inversion results thanks to the computational gain is promising, and hopefully will motivate further research on the intersections of underwater geoacoustic inversion problems and ML/DL.

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